FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 1HA-I 2208 — D(7081)

Retyped faithfully from the original board paper — same questions, no changes.

Note: PHYSICS HSSC–I. SECTION – A (Marks 17). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7081. Fill the relevant bubble against each question:

Q1. Which of the following pairs has same dimension?

(A) Pressure, Density
(B) Impulse, Momentum
(C) Stress, Strain
(D) Momentum, Inertia

Q2. The number of significant figures in 0.000125010 are:

(A) 3
(B) 4
(C) 5
(D) 6

Q3. Two forces of magnitudes F_1 and F_2 acting at right angle to each other have the resultant of the magnitude:

(A) $(F_1+F_2)/2$
(B) $F_1^2+F_2^2$
(C) $\sqrt{(F_1^2+F_2^2)}$
(D) $(F_1^2+F_2^2)/2$

Q4. The distance covered by a body in time t, starting from rest is:

(A) at^2
(B) $2at^2$
(C) a^2t
(D) $\frac{1}{2}at^2$

Q5. The horizontal range of projectile is same for the angles:

(A) 30° and 40°
(B) 40° and 50°
(C) 60° and 70°
(D) 80° and 90°

Q6. A ball of mass 100g is thrown vertically upward at a speed of $25ms^{-1}$. If no energy is lost, determine the height it would reach. (Loss in K.E=Gain in P.E)

(A) 31.9m
(B) 1.28m
(C) 63.78m
(D) 321.5m

Q7. The mass of a body is m, its speed is v and K.E is E. When mass is doubled and its speed is reduced to half, then K.E will be:

(A) 2E
(B) E/2
(C) 4E
(D) E/4

Q8. The angular displacement of one radian is:

(A) 47.3°
(B) 57.3°
(C) 67.3°
(D) 77.3°

Q9. The ratio of the linear velocities of the points at distances r and r/4 from the axis of rotation of a rigid body is:

(A) 0.25
(B) 0.5
(C) 2
(D) 4

Q10. Two rain drops have radii in the ratio 2:3. The ratio between their terminal velocities will be:

(A) 2:3
(B) 3:2
(C) 4:9
(D) 9:4

Q11. The length of a second pendulum is:

(A) 70cm
(B) 80cm
(C) 90cm
(D) 100cm

Q12. When amplitude of a wave becomes double, its energy becomes:

(A) 2 times
(B) ½ times
(C) 4 times
(D) ¼ times

Q13. According to Laplace correction, sound travels in air under the condition of:

(A) Isothermal process
(B) Adiabatic process
(C) Isochoric process
(D) Isobaric process

Q14. The velocity of sound in air would become double to its velocity at 0°C at temperature:

(A) 313°C
(B) 586°C
(C) 819°C
(D) 1172°C

Q15. Fringe spacing =

(A) Lλ/D
(B) Dλ/L
(C) λ/(DL)
(D) L/(λD)

Q16. According to first law of thermodynamics, Which one is correct?

(A) Cp+Cv = R
(B) Cp = 1+R/Cv
(C) R = Cv/Cp
(D) Cp = R+Cv

Q17. A Carnot engine works between ice point and steam point. Its efficiency will be:

(A) 26.81%
(B) 53.36%
(C) 62.46%
(D) 71.23%

> Note: Important formulae: P = F/A; Density = M/V; P = mv; δ = F/A; ε = ΔL/L; J = F×Δt; v_t = v_o + (0.61)t; T = 2π√(l/g); P.E = mgh; v_t/v_o = √(T/T_o); g = 9.8ms⁻²; R = vi²sin(2θ)/g; %Efficiency = ((T₁−T₂)/T₁)100%; S = rθ; ω = θ/t; K.E = ½mv²; v_o = 332ms⁻¹ at 0°C; S = v_i t + ½at²; 2π radians = 360°; V_terminal = 2ρgr²/(9η). —1HA-I 2208-3081 (HA)—. Page 1 of 2. Page 2 of 2.

> Note: PHYSICS HSSC–I. Time allowed: 2:35 Hours. Total Marks Sections B and C: 68. NOTE: Answer any FOURTEEN parts from Section 'B' and attempts any TWO questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly.

Q. SECTION – B (Marks 42)

Q. 2 Answer any FOURTEEN parts. All parts carry equal marks. (14 x 3 = 42)

(i) Briefly describe necessary conditions for SHM.

(ii) What is torque? Define torque as vector product of r and F.

(iii) For vector A show that |A| = √(Aₓ²+A_y²)

(iv) Find the change in momentum for an object subjected to a given force for a given time and state law of motion in terms of momentum.

(v) Two balls are projected in directions at 15° and 45° with the horizontal. If both attained the same range then find the ratio of their initial speeds.

(vi) Calculate the work done in kilojoules in lifting a mass of 10kg at a steady velocity through a vertical height of 10m?

(vii) If radius of moon is 1/6 times radius of earth and gravity on moon is 1/5 times gravity on earth, then find the escape velocity at the surface of moon?

(viii) Show that angular momentum L = mvr

(ix) Find the relation between linear velocity and angular velocity.

(x) Discuss working principle of aerofoil.

(xi) Show that in SHM, the acceleration is zero when the velocity is greatest and the velocity is zero when the acceleration is greatest.

(xii) Discuss the interchanging between K.E. and P.E during SHM.

(xiii) As a result of distant explosion, an observer senses a ground tremor and then hears the explosion. Explain the time difference in it.

(xiv) The speed of sound in air at 0°C is 332ms⁻¹. What will be its speed at 25°C?

(xv) Under what conditions two or more sources of light behave as coherent sources?

(xvi) In Young's double slit experiment, if the distance between the slits is halved and distance between slit and screen is doubled, then find the change in fringe width?

(xvii) A garden hose of inner radius 1.25cm carries water at $2.60 ms^{-1}$. The nozzle at the end has radius 0.30cm. How fast does the water emerge out through the nozzle?

(xviii) Show that $n_2/n_1 = \tan i_p$ (polarization of transverse waves)

(xix) Is it possible to convert internal energy (ΔU) into mechanical energy? Explain with an example.

(xx) Describe the terms 'specific heat' and 'molar specific heat' of gases.

Q. SECTION – C (Marks 26)

Note: Attempt any TWO questions. All questions carry equal marks. (2 x 13 = 26)

Q. 3 a. Explain vector and scalar products of two vectors with neat diagrams. (05) b. Describe two conditions of equilibrium. (04) c. What are the dimensions and units of gravitational constant G in the formula $F = Gm_1m_2/r^2$? (04)

Q. 4 a. Explain and derive a mathematical relation for Absolute Potential Energy. (05) b. What is meant by moment of inertia of a body? Derive a formula for it. (04) c. What is the aero foils lift (in newtons) on a wing of area $88 m^2$ if the air passes at speed over its top surface at $280 ms^{-1}$ and bottom surface at $150 ms^{-1}$? (04)

Q. 5 a. Explain Doppler's effect in detail with its special cases. (05) b. The radius of sphere 'r' is measured with a Vernier Callipers as $(r \pm \Delta r) = (2.25 \pm 0.01)cm$. Calculate the volume of sphere. (04) c. A Carnot engine utilizes an ideal gas. The source temperature is 227°C and sink temperature is 127°C. Find the efficiency of the engine. Also find heat input from the source and heat rejected to the sink when 10000J of work is done? (04)

> Note: Important formulae: Fringe Spacing = $L\lambda/D$; $R = v_i^2 \sin(2\theta)/g$; Work = $F \cdot d$; $v_t = v_o + (0.61)t$; $v = r\omega$; $v_{esc} = \sqrt{(2gR)}$; Weight = mg; $v_o = x_o\sqrt{(k/m)}$; %Efficiency = $((T_1 - T_2)/T_1) \times 100\% = ((Q_1 - Q_2)/Q_1) \times 100\%$. —1HA-I 2208 (HA)—. Page 1 of 1 (Physics).